

PREPARATION OF YEAST DNA, RNA, AND PROTEINS

SECTION IV

Preparation of Yeast DNA

UNIT 13.11

Molecular studies in yeast often require the isolation of both plasmid and chromosomal yeast DNA. Plasmid DNA is used in the transformation of *E. coli*, whereas chromosomal DNA is used for Southern hybridization analysis, in vitro amplification by the polymerase chain reaction (PCR), or cloning of integrated plasmids. This unit presents two variations of the “smash and grab” protocol (Hoffman and Winston, 1987) that produce suitable DNA for all these applications. These protocols work for both *Saccharomyces cerevisiae* and *Schizosaccharomyces pombe*, although for *S. pombe* a suitable liquid medium must be used to grow the cells.

RAPID ISOLATION OF PLASMID DNA FROM YEAST

BASIC PROTOCOL

Plasmid DNA is released from a yeast transformant along with chromosomal DNA in a rapid, 10-min protocol by vortexing with glass beads in the presence of detergents, phenol, chloroform, and isoamyl alcohol. After centrifugation to pellet cell debris, no further purification of the DNA is required for transformation of competent *E. coli* cells.

Materials

- YPD or appropriate selective medium (UNIT 13.1)
- Yeast colony containing the plasmid of interest
- Breaking buffer (see recipe)
- 25:24:1 (v/v/v) phenol/chloroform/isoamyl alcohol (with buffered phenol; UNIT 2.1)
- 0.45- to 0.52-mm acid-washed glass beads (UNIT 13.12; Thomas Scientific)
- Competent *E. coli* cells HB101 or MH1 (UNIT 1.8)
- LB plates (UNIT 1.1) containing appropriate antibiotic (UNIT 1.4)
- 13 × 100-mm glass tubes, sterile
- 30°C incubator with shaker or roller drum
- Additional reagents and equipment for growth of *E. coli* on solid medium (UNIT 1.3) and transformation of *E. coli* (UNIT 1.8)

1. Inoculate 2 ml medium in a 13 × 100-mm sterile glass tube with a single yeast colony containing the plasmid of interest. Grow overnight to stationary phase at 30°C in either a roller drum or a shaking incubator.

Transformants carrying stable plasmids (UNIT 13.4) should be grown in YPD to get a dense culture. Transformants carrying unstable plasmids should be grown in selective medium to increase the percentage of plasmid-containing cells.

2. Transfer 1.5 ml of the overnight culture to a microcentrifuge tube and spin 5 sec at high speed, room temperature. Pour off supernatant and disrupt pellet by vortexing briefly.
3. Resuspend cells in 200 µl breaking buffer. Add 0.3 g glass beads (~200 µl volume) and 200 µl phenol/chloroform/isoamyl alcohol. Vortex 2 min at highest speed.

The amount of vortexing required can vary depending upon the vortex used. Determine by microscopic examination the minimum vortexing required for a particular machine to break 80% to 90% of the cells.

For simultaneous preparation of DNA from several yeast strains, a multi-tube vortexer or a multi-tube head for a standard vortexer can be used (VWR Scientific) and vortexing time should be increased to 3 min.

*Saccharomyces
cerevisiae*

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**ALTERNATE
PROTOCOL**

4. Microcentrifuge 5 min at high speed, room temperature.
5. Transform competent *E. coli* HB101 or MH1 with 1 to 2 μ l of the aqueous layer. Plate on LB plates containing the appropriate antibiotic to select for the drug-resistance marker on the plasmid.

See Critical Parameters and Troubleshooting for guidelines on transformation.

Save 50 μ l of the aqueous layer and store at -20°C in case additional transformations are required.

RAPID ISOLATION OF YEAST CHROMOSOMAL DNA

The DNA preparation described in the Basic Protocol can be easily scaled up to prepare chromosomal DNA for use in Southern hybridization analysis (UNIT 2.9A), in vitro DNA amplification by PCR (UNIT 15.1), or restriction digestion and ligation (UNITS 3.1 & 3.16) to clone integrated plasmids. This procedure is significantly faster than other protocols used to isolate high-molecular-weight DNA. Although the DNA is subject to shearing in this protocol, the resulting DNA is of sufficiently high molecular weight to enable detection of 19-kb restriction digestion products by Southern hybridization analysis.

Additional Materials (also see Basic Protocol)

- TE buffer (APPENDIX 2)
- 1 mg/ml DNase-free RNase A (UNIT 5.5)
- 4 M ammonium acetate solution (APPENDIX 2)
- 100% ethanol
- 18 $\times$ 150-mm glass culture tubes or 17 $\times$ 100-mm disposable polypropylene tubes, sterile
- Tabletop centrifuge

Prepare yeast cells

1. Grow a 10-ml culture of yeast in YPD overnight to stationary phase in either 18 $\times$ 150-mm sterile glass culture tubes or in 17 $\times$ 100-mm sterile, polypropylene tubes (see Basic Protocol, step 1).
2. Spin culture 5 min in a tabletop centrifuge at $1200 \times g$ (3000 rpm), room temperature. Aspirate or pour off supernatant, and resuspend cells in 0.5 ml water.
3. Transfer the resuspended cells to a microcentrifuge tube and spin 5 sec at room temperature. Pour off supernatant and disrupt pellet by vortexing briefly.

This wash step in water removes any remaining medium.

Break open the cells

4. Resuspend cells in 200 μ l breaking buffer. Add 0.3 g glass beads (~200 μ l volume) and 200 μ l phenol/chloroform/isoamyl alcohol and vortex at highest speed for 3 min.

The amount of vortexing required can vary depending upon the vortex used. Determine by microscopic examination the minimum vortexing required for a particular machine to break 80% to 90% of the cells.

If using a multi-tube vortex rather than a single-tube vortex, the time of vortexing may need to be increased up to 4 or 5 min to get more efficient breakage of cells. However, longer vortexing can result in shearing of DNA molecules.

5. Add 200 μ l TE buffer and vortex briefly.
6. Microcentrifuge 5 min at high speed, room temperature, and transfer aqueous layer to a clean microcentrifuge tube. Add 1 ml of 100% ethanol and mix by inversion.

7. Microcentrifuge 3 min at high speed, room temperature. Remove supernatant and resuspend pellet in 0.4 ml TE buffer.

Degrade RNA contaminants and recover DNA

8. Add 30 μ l of 1 mg/ml DNase-free RNase A, mix, and incubate 5 min at 37°C.
9. Add 10 μ l of 4 M ammonium acetate and 1 ml of 100% ethanol. Mix by inversion.
10. Microcentrifuge 3 min at high speed, room temperature. Discard supernatant and dry pellet. Resuspend DNA in 100 μ l TE buffer.

Yields of ~20 μ g of chromosomal DNA should be obtained. This DNA is ready to use for restriction digestion (UNIT 3.1), in vitro PCR amplification (UNIT 15.1), or Southern blot analysis (UNIT 2.9). For Southern blots, best results are obtained when 5 μ l DNA (~1 μ g) is digested in a total volume of 20 μ l. To amplify by PCR, 2 μ l of DNA should be used in a 50- μ l reaction.

REAGENTS AND SOLUTIONS

Use deionized, distilled water in all recipes and protocol steps. For common stock solutions, see APPENDIX 2; for suppliers, see APPENDIX 4.

Breaking buffer

2% (v/v) Triton X-100
1% (v/v) sodium dodecyl sulfate (SDS)
100 mM NaCl
10 mM Tris-Cl, pH 8.0
1 mM EDTA, pH 8.0
Store $\leq$ 1 year at room temperature

COMMENTARY

Background Information

These DNA isolation protocols simultaneously break yeast cells by vortexing with glass beads in a detergent solution and separate nucleic acids from protein by phenol/chloroform extraction. They differ in the amount of additional manipulation required to obtain DNA of the quality needed for a particular procedure. Other protocols also describe the isolation of high-molecular-weight DNA (Cryer et al., 1975; Winston et al., 1983; Davis et al., 1980). However, even the protocols designed to be rapid DNA preparations require >3 hr time. In general, the protocols described here produce DNA of sufficient molecular weight for any desired application, except for the production of insert DNA for the construction of genomic libraries.

Many situations require the isolation of plasmid DNA from a yeast transformant to transform *E. coli*. Examples include cloning of (1) the wild-type copy of a gene by complementation of a recessive mutant (UNIT 13.8), (2) a dominant mutant allele of a gene by transformation of a wild-type strain, or (3) a mutant allele of a gene by plasmid gap repair (UNIT 13.10). Plasmid DNA is also needed to identify a de-

sired mutant allele of a gene by the plasmid shuffle technique (UNIT 13.10). Most yeast plasmids used in molecular biology are shuttle vectors carrying a selectable marker and an origin of replication (*ori*) for transformation of and maintenance in *E. coli* (UNIT 13.4). Because significantly more plasmid DNA can be isolated from an *E. coli* transformant than from a yeast transformant, *E. coli* is used to amplify the DNA for restriction mapping and other in vitro manipulations of the DNA as well as for retransformation of yeast strains.

Other situations require the isolation of chromosomal DNA. The most common use of this DNA is for Southern hybridization analyses. Such analyses are needed to confirm that integration of a YIp plasmid or of a marked gene disruption has occurred by homologous recombination. Chromosomal DNA can also be used as a template for in vitro amplification by PCR in place of Southern analyses to determine whether DNA is integrated by homologous recombination. PCR can also be used in place of gap repair to clone mutant alleles of a given gene. Finally, chromosomal DNA is needed to clone DNA adjacent to an integrated plasmid. This is done by digesting the chromosomal

Saccharomyces cerevisiae

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DNA with a restriction enzyme that creates a restriction fragment containing the plasmid origin of replication and selectable marker, as well as the flanking DNA of interest (Southern analyses are required to determine the appropriate restriction enzyme for this construction). Ligation of this yeast DNA, followed by transformation of *E. coli*, results in the isolation of the cloned flanking DNA, as only the appropriate piece of DNA will confer drug resistance upon the *E. coli* strain.

These protocols do not produce intact chromosomal DNA. Yeast chromosomes can be isolated by methods involving gentle lysis of cells in agarose blocks (UNIT 2.5B; Carle and Olson, 1987). Intact chromosomes can be used to map a cloned gene to a given chromosome by Southern hybridization analysis or as a size standard for field-inversion or orthogonal-field-alteration gel electrophoresis (UNIT 2.5B).

Critical Parameters and Troubleshooting

There is a significant difference among *E. coli* strains regarding their transformation efficiency with plasmid DNA isolated by this protocol. One commonly used strain, HB101 (Table 1.4.5), works well with either CaCl_2 transformation or electroporation protocols (UNIT 1.8). A less commonly used strain, MH1 (Hall et al., 1984), is even better suited for these transformations when made competent by the RbCl method (Hanahan, 1983). Because MH1 is a derivative of MC1061, it is possible that this strain would also transform well using the Hanahan protocol. Many other strains do not work as well, even when they appear to be more competent as judged by transformation with a control plasmid used to standardize transformation efficiency. Therefore, problems with transformations may be due to the choice of the *E. coli* host strain.

It is important to note that only 1 or 2 μl of the aqueous layer should be used per transformation, as larger amounts are increasingly toxic to *E. coli*. Attempts to purify the DNA with ethanol precipitations are generally unsuccessful, possibly due to co-precipitation of material that inhibits transformation. A 1:1 isopropanol precipitation can be effective in concentrating the DNA when the original attempt fails to produce transformants. In general, however, this additional manipulation is unwarranted.

When preparing DNA for use in Southern hybridization analysis, it is important to minimize the vortexing time needed to break the cells in order to reduce shearing of the DNA.

Three minutes is generally sufficient, but this time can vary from vortex to vortex. Too much vortexing can result in a weak signal for restriction fragments <10 kb. The other consideration for optimizing Southern blot analysis is the amount of DNA used per digest. This protocol generates a very consistent yield of readily digested DNA. Increasing the amount of DNA per digest can result in incomplete cutting and is not recommended.

Anticipated Results

The basic protocol for isolating plasmid DNA from yeast transformants gives sufficient DNA for transformation of *E. coli*. The number of transformants obtained is a function of copy number of plasmid and transformation efficiency of *E. coli* host strain. The protocol for isolating chromosomal DNA yields ~20 μg DNA from a 10-ml stationary-phase culture.

Time Considerations

It should take ~10 min to prepare plasmid DNA for transformation of *E. coli*. The genomic DNA preparation takes from 45 min to 1 hr for twelve cultures. The vortexing step can become much more time-consuming if only a single-tube vortex is available, as only three microcentrifuge tubes can be vortexed simultaneously.

Literature Cited

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